

REED PUTNAM PRACTICE · FALL 2026

Linear Algebra: Determinants, Eigenvalues & Rank

Week 7 · turn a matrix statement into a scalar one

Thu Nov 12 / Fri Nov 13

The toolbox (1)

Determinant

Multilinear and **alternating** in the rows: row operations are free, a swap flips the sign, equal rows give 0. $\det(AB) = \det A \det B$, $\det A^T = \det A$, block-triangular $\Rightarrow$ product of the blocks, and $\det A = \prod \lambda_i$ over the complex eigenvalues with multiplicity.

- **Eigenvalues & trace:** $\operatorname{tr} A = \sum \lambda_i$, $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. If $p(A) = 0$, every eigenvalue is a root of p . Real matrix $\Rightarrow$ nonreal eigenvalues come in **conjugate pairs**, $\lambda \bar{\lambda} = |\lambda|^2 > 0$.
- **Rank:** $\operatorname{rank} A + \dim \ker A = n$; $\det A \neq 0 \iff \ker A = 0$. $\operatorname{rank}(M - \lambda I) = r \Rightarrow \lambda$ has multiplicity $\geq n - r$, the trace pins down the rest.

The toolbox (2)

A determinant with a parameter is a polynomial

Find its **roots** (where two rows coincide) and its **degree**; one leading coefficient finishes it. Vandermonde: $\det[x_i^{j-1}] = \prod_{i < j} (x_j - x_i)$.

- **Structured matrices**: rank one uv^T has $\det(I + uv^T) = 1 + v^T u$; all-ones J has eigenvalues n and 0 ; entries in $\min(i, j)$ or $|i - j|$: **subtract consecutive rows**.
- **Finite fields**: reduction mod p commutes with $\det$, so an integer matrix has $\det \not\equiv 0 \pmod{p}$ iff it is invertible over $\mathbb{F}_p$. Over $\mathbb{F}_2$ **signs vanish**. $|GL_n(\mathbb{F}_q)| = \prod_{i=0}^{n-1} (q^n - q^i)$.

Example 1

Rank: a one-sided inverse is two-sided

If A, B are $n \times n$ matrices and $AB = I$, then $BA = I$.

→ work it on the board (~3 min)

Example 2

A determinant is a polynomial: Vandermonde 3×3

Compute $V = \det \begin{pmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{pmatrix}$.

→ work it on the board (~3 min)

Example 3

Eigenvalues: a matrix equation forces a sign

If A is a real $n \times n$ matrix with $A^3 = A + I$, then $\det A > 0$.

→ work it on the board (~3 min)

How to recognize it

- > A matrix in the statement: pick the right **scalar invariant**.
- > “No such matrices,” “ n must be even”: **det or tr of both sides**.
- > “Compute det”: rank one, row differences, blocks, polynomial in a parameter; **do $n = 1, 2, 3$ first**.
- > “ $A^k = I$,” “ $A^2 = A$ ”: eigenvalues obey the same equation; conjugate pairs give **parity**.
- > Odd det, 0/1 or integer entries: work over $\mathbb{F}_2$. “Independent,” “injective”: **rank**.

Pitfalls

- > $\det = \prod \lambda_i$, $\text{tr} = \sum \lambda_i$: complex eigenvalues, algebraic multiplicity; not real, distinct, or diagonalizable unless you say why.
- > $\text{tr}(ABC) = \text{tr}(BCA) \neq \text{tr}(BAC)$; $AB = 0 \not\Rightarrow A = 0$ or $B = 0$; over $\mathbb{F}_p$, $\text{tr } I = n$ can be 0.

Now you

Problems 1–4.

1–2 warm-ups · 3–4 easy Putnam · 5–6 take-home

35 minutes, groups of 2–3

Solutions posted after Friday.